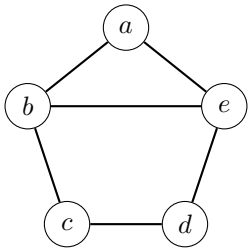
The Strong Chromatic Index and the Erdős–Nešetřil Conjecture

The setup, with pictures (start here if you need it)

Graphs: dots and lines.

A *graph* G is a finite set of *vertices* (dots) together with a set of *edges* (lines) joining some pairs of vertices. That is it. No coordinates, no geometry — just "which dots are joined by a line." Here is a graph with 5 vertices and 6 edges:



Degree, and what $\Delta(G)$ means.

The *degree* of a vertex v, written $\deg(v)$, is the number of edges that touch it. Just count the lines coming out of the dot. For the graph above:

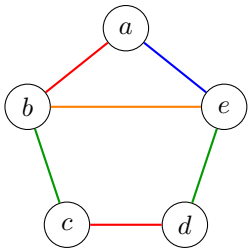
vertex	a	b	c	d	e
deg	2	3	2	2	3

The **maximum degree** of G, written $\Delta(G)$, is the largest degree of any vertex in G. For our example, $\Delta(G) = 3$ (achieved by b and e). That is the single most important number describing a graph for our purposes.

In words: "$\Delta(G) = k$" means *every vertex has at most k edges sticking out of it, and at least one vertex has exactly k.*

Edge colouring: the easy version.

A *proper edge colouring* is an assignment of a colour to every edge such that **any two edges meeting at the same vertex get different colours**. That is the only rule.



a proper edge colouring with 4 colours

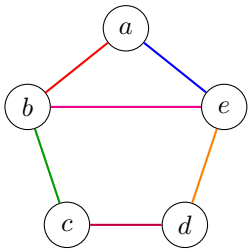
The minimum number of colours that suffices is the *chromatic index* $\chi'(G)$. Vizing's theorem (1964) says $\chi'(G)$ is either $\Delta(G)$ or $\Delta(G)+1$. Cleanly bounded by $\Delta(G)$.

Strong edge colouring: the hard version.

A *strong edge colouring* adds a second rule. Now:

- Edges meeting at a vertex must get different colours (as before), *and*
- Edges that are **joined by another edge** must *also* get different colours.

Equivalently: every colour class must be an *induced matching* — a set of edges with no two of them within "one hop" of each other.



strong colouring: every two edges within 1 hop differ

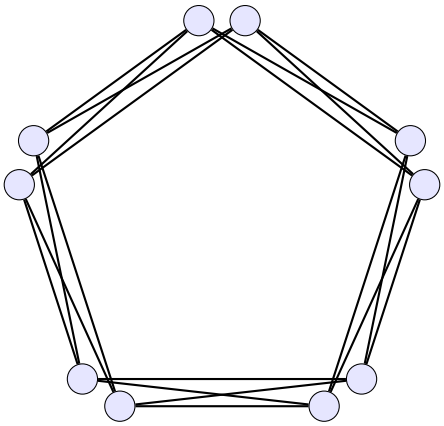
For the graph above, the strong rule forces 6 different colours (every pair of edges is within one hop of each other, because the graph is small). So $\chi'_s(G) = 6$ here.

The minimum number of colours that makes a strong colouring possible is the *strong chromatic index* $\chi'_s(G)$. This is the quantity the rest of the paper is about.

One more picture: why χ'_s blows up.

Even though ordinary $\chi'(G)$ is at most $\Delta(G)+1$, the strong version $\chi'_s(G)$ can be *quadratic* in $\Delta(G)$ — much bigger. The reason is the *blow-up of* C_5, which we will keep returning to.

$C_5[2]$**:** take the 5-cycle, replace each vertex with 2 twin vertices, and join twins across the cycle. The result has 10 vertices, 20 edges, every vertex has degree $\Delta = 4$, and *every pair of edges is within* 1 *hop of each other*. So every edge needs its own colour:



So $\chi'_s(C_5[2]) = 20 = 5 \cdot 4^2/4$. This is the conjectured worst case for $\Delta = 4$. The Erdős–Nešetřil conjecture says: *no graph with $\Delta = k$ ever needs more than $5k^2/4$ colours for a strong edge colouring.*

That is the conjecture. Everything after this section is about trying to prove it.

Detour: a proof of Vizing's theorem

Before diving into the *strong* version, here is a self-contained proof of the ordinary version. It is short, beautiful, and worth seeing once.

Theorem (Vizing, 1964)**.** *For any simple graph G,*

$$\chi'(G) \;\leq\; \Delta(G) + 1.$$

The lower bound $\chi'(G) \geq \Delta(G)$ is trivial: the $\Delta(G)$ edges meeting a max-degree vertex must all receive different colours. The interesting part is the upper bound.

Proof. Set $k = \Delta(G) + 1$. We show G admits a proper edge colouring with k colours, by induction on $|E(G)|$. The base case $|E(G)| = 0$ is trivial.

Let $e = xy_1$ be any edge of G. By induction, $G - e$ has a proper edge colouring c using colours from $\{1, 2, \ldots, k\}$. We will either extend c to colour e, or perturb c on a few edges so that we can.

For each vertex v, let $M(v) \subseteq \{1, \ldots, k\}$ be the set of colours not appearing on any edge incident to v in $G - e$. Since $\deg_G(v) \leq \Delta(G) = k - 1$, we have $|M(v)| \geq 1$.

Easy case. If $M(x) \cap M(y_1) \neq \emptyset$, pick any colour in the intersection and use it on e. Done.

The fan. Otherwise, build a sequence $y_1, y_2, y_3, \ldots$ of distinct neighbours of x as follows. Given y_i, pick any colour $\beta_i \in M(y_i)$. If β_i does not appear on any edge at x, the fan stops (*type A* stop). Otherwise, β_i appears on a unique edge xy_{i+1} at x. Set y_{i+1} accordingly and continue.

Because the colours β_i are pairwise distinct (each is the colour of a distinct edge at x) and there are only k colours, the fan must stop. Two types of stops are possible:

Type A: $\beta_t \in M(x)$**.** "Shift" the colours along the fan. New colouring c':

$$c'(xy_i) \;=\; \beta_i \quad \text{for } i = 1, 2, \ldots, t,$$

all other edges unchanged. Each $\beta_i \in M(y_i)$, so there is no conflict at y_i. At x the new colours $\beta_1, \ldots, \beta_t$ are distinct, $\beta_t \in M(x)$, and the β_i for $i < t$ are colours of edges that used to be at x, shifted by one. A direct check shows c' is proper.

Type B: $\beta_t = \beta_j$ for some $j < t$. (The colour missing at the last fan-tip already appears earlier in the fan.) Now invoke a *Kempe-chain swap*. Pick any colour $\alpha \in M(x)$ (exists because $|M(x)| \geq 1$, and $\alpha \neq \beta_j$ because β_j appears at x on edge xy_{j+1}).

Consider the subgraph H of $G - e$ consisting of all edges coloured α or β_j. Every vertex has degree ≤ 2 in H, so H is a disjoint union of paths and even cycles. The component P containing y_t is a path (because $\beta_j \in M(y_t)$ so y_t has degree at most 1 in H). Swap the two colours along P to obtain a new proper colouring c'' of $G - e$.

Where does P end? It cannot end at x on the α-side (because $\alpha \in M(x)$). It cannot pass through x on the β_j-side via xy_{j+1} except at an endpoint. A careful case analysis (split on whether P contains y_j or not) shows that after the swap, the fan $y_1, \ldots, y_j$ now ends in a *type-A* stop: either $\alpha \in M(x) \cap M(y_j)$ in the post-swap colouring, or some prefix of the fan satisfies type A. Apply the shifting argument from type A to that fan, in the swapped colouring. The result is a proper k-colouring of G. □

Remarks.

- The proof is constructive and gives a polynomial-time algorithm: build the fan, decide type, shift or swap. Each step touches $O(|V|)$ edges, and recolouring an edge can be amortised.
- Vizing's bound is tight: the Petersen graph has $\Delta = 3$ and $\chi' = 4$. Graphs with $\chi' = \Delta + 1$ are called *class 2*; the others are *class 1*. Distinguishing the two classes is NP-complete (Holyer, 1981) — so even though the bound is clean, deciding equality is hard.
- For the *strong* chromatic index, no analogue of Vizing's theorem is known (or expected): the strong version can grow quadratically in Δ, which is why the next section is hard.

From edge colouring to strong edge colouring

A *proper edge colouring* of a graph G assigns colours to the edges so that any two edges sharing a vertex get different colours. The minimum number of colours needed is the *chromatic index* $\chi'(G)$. As just proved,

$$\Delta(G) \;\leq\; \chi'(G) \;\leq\; \Delta(G) + 1,$$

so the chromatic index is essentially determined by the maximum degree. Tight, clean, well-understood.

Strong edge colouring tightens the requirement: not only must two edges sharing a vertex get different colours, but two edges *joined by an edge* must also get different colours. Equivalently:

> A **strong edge colouring** is a proper edge colouring in which every colour class is an *induced matching*.

The minimum number of colours needed is the *strong chromatic index* $\chi'_s(G)$. Equivalently, $\chi'_s(G) = \chi(L(G)^2)$, the chromatic number of the square of the line graph: two edges of G get different colours iff they are at distance at most 2 in $L(G)$.

Unlike the chromatic index, the strong chromatic index is *not* controlled tightly by Δ. The trivial bound is

$$\chi'_s(G) \;\le\; 2\Delta(G)^2 - 2\Delta(G) + 1,$$

proved by greedy colouring (each edge has at most $2\Delta - 2$ adjacent edges, each of which has at most $2\Delta - 2$ further edges incident, plus itself). The interesting question is how much the leading constant 2 can be reduced.

A little history

The strong chromatic index sits at the end of a hundred-year arc of edge-colouring results. The shape of that arc explains why the Erdős–Nešetřil conjecture looks like it does, and why the constant 5/4 feels so resistant.

1916 — Kőnig

Dénes Kőnig, working in Budapest, proves that the chromatic index of a bipartite graph equals its maximum degree:

$$\chi'(G) \;=\; \Delta(G) \quad \text{if } G \text{ is bipartite.}$$

The proof is via augmenting paths, the same combinatorial idea that would later underpin matching theory and network flow. Kőnig's theorem is the first edge-colouring theorem, and it sets the tone: the chromatic index is, in nice classes, exactly the obvious lower bound.

1949 — Shannon

Claude Shannon, in the same year as his foundational paper on information theory, proves that for *multigraphs* (graphs with parallel edges allowed),

$$\chi'(G) \;\le\; \lfloor 3\Delta(G)/2 \rfloor.$$

The factor 3/2 is tight, attained by the multigraph on three vertices with $\Delta/2$ parallel edges between each pair. Shannon's motivation came from connection scheduling in telephone networks.

1964 — Vizing

Vadim Vizing, then a junior researcher at the Novosibirsk Institute of Mathematics, proves the theorem with his name on it: for every simple graph,

$$\chi'(G) \;\le\; \Delta(G) + 1.$$

The proof is the fan-and-Kempe-chain argument given in the previous section. The paper appears in *Diskretnyĭ Analiz*, a relatively obscure Soviet journal; it takes several years for the result to become widely known in the West.

Independently and at roughly the same time, Ram Prakash Gupta proves the same result in his 1966 doctoral work in the United States. The theorem is sometimes called the Vizing–Gupta theorem in older literature.

Vizing's theorem is fundamental, but it leaves a question: which graphs need $\Delta + 1$ colours and which only Δ?

1981 — Holyer's hardness result

Ian Holyer shows that deciding whether $\chi'(G) = \Delta(G)$ (i.e. whether a graph is "class 1") is NP-complete, even restricted to cubic graphs. So the Vizing bound is sharp in the strongest possible sense: you cannot improve it by an efficient algorithm.

1983 — Fouquet and Jolivet name the strong version

The first paper to study strong edge colouring as its own object is due to Jean-Luc Fouquet and Jean-Luc Jolivet: "Strong edge-colourings of graphs and applications to multi-k-gons," *Ars Combinatoria* 16*A* (1983). They conjecture the cubic case: $\chi'_s(G) \le 10$ for $\Delta(G) \le 3$, with the Petersen graph as motivating example. This is the first appearance of the bound $5\Delta^2/4$ for any specific Δ.

1985 — Erdős and Nešetřil in Prague

Paul Erdős and Jaroslav Nešetřil announce the general conjecture during a meeting in Prague. By this time Erdős, 72, has been pseudo-domiciled in Hungary and on a permanent visiting tour through North America, Europe and Israel for years; Nešetřil, 39, is at Charles University and a frequent host. Their collaboration on Ramsey-type combinatorics is by then over a decade old. The conjecture is recorded in Erdős's 1988 survey, "Problems and results in combinatorial analysis and graph theory" (*Discrete Mathematics* 72, 81–92).

The conjecture is, characteristically for Erdős, stated cleanly, believed to be sharp, and accompanied by no proof. The C_5-blow-up is named on the spot as the family forcing $5k^2/4$.

1992 — Andersen and the cubic case

L. D. Andersen, working in Aalborg, proves the Fouquet–Jolivet conjecture: $\chi'_s(G) \le 10$ for $\Delta(G) \le 3$. The proof is a careful structural reduction on a minimum counter-example. Almost simultaneously, Horák, Qing and Trotter publish an independent proof using induced-matching language. Both papers were submitted in 1990.

This is the first — and so far the only — complete case of the Erdős–Nešetřil conjecture.

1997 — Molloy, Reed, and the local lemma

Michael Molloy and Bruce Reed, building on the constructive uses of the *Lovász Local Lemma* pioneered by Beck (1991) and Alon (1991), prove the first nontrivial general bound: $\chi'_s(G) \le 1.998\,\Delta^2$ for Δ large.

The local lemma had been around since 1975, but it was the 1990s explosion of probabilistic graph colouring (culminating in Molloy and Reed's book *Graph Colouring and the Probabilistic Method*, 2002) that made constants below 2 possible. This is where the modern story begins: every subsequent advance — Bruhn–Joos, Bonamy–Perrett–Postle, Hurley–Kang — is a refinement of the same template.

1996 — Erdős dies, the problem stays

Paul Erdős dies in Warsaw in September 1996, age 83, leaving behind hundreds of unsolved problems. The strong-chromatic-index conjecture is one of the cleaner ones: short to state, deeply

unresolved, with a single extremal family clearly visible from a hundred metres away. It is the kind of problem he liked to say should "not be hard, but is hard."

2006–**present**

Cranston's $\chi'_s \le 22$ at $\Delta = 4$ (2006); Huang–Santana–Yu $\chi'_s \le 21$ at $\Delta = 4$ (2018); Bruhn–Joos 1.93 (2018); Bonamy–Perrett–Postle 1.835 (2018); Hurley–de Joannis de Verclos–Kang 1.772 (2021). Each advance moves the constant a little. The C_5-blow-up still stands as the only known extremal family. The constant $5/4$ has not budged.

The arc, in one line: Kőnig → Shannon → Vizing → Fouquet–Jolivet → Erdős–Nešetřil → Andersen → Molloy–Reed. The conjecture is the unresolved endpoint of a sequence of theorems that get cleaner the further back you go.

The conjecture

In 1985, at a meeting in Prague, Paul Erdős and Jaroslav Nešetřil announced the bound they believed was the truth:

Conjecture (Erdős–Nešetřil, 1985)**.** *Let G be a graph with maximum degree $\Delta(G) = k$. Then*

$$\chi'_s(G) \;\le\; \begin{cases} \dfrac{5k^2}{4} & \text{if } k \text{ is even,} \\[2ex] \dfrac{5k^2 - 2k + 1}{4} & \text{if } k \text{ is odd.} \end{cases}$$

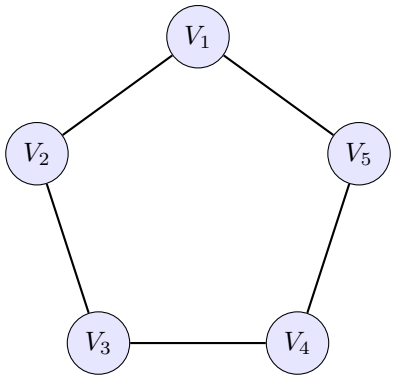
The odd-case formula is exactly $\frac{5k^2}{4} - \frac{k}{2} + \frac{1}{4}$. For small k:

k	conjectured χ'_s	note
2	5	matched by C_5
3	10	proved by Andersen (1992)
4	20	open; best known 22
5	29	open
6	45	open
7	58	open

Why $5k^2/4$? The blow-up of C_5

The constant $5/4$ is not a guess. It is exactly what a single family of graphs forces.

Construction. Take the 5-cycle C_5 with vertices v_1, v_2, v_3, v_4, v_5. Replace each v_i with an independent set V_i of size $\lceil k/2 \rceil$ or $\lfloor k/2 \rfloor$ (chosen around the cycle so the result is k-regular). Join $u \in V_i$ to $w \in V_j$ iff $v_i v_j$ was an edge of C_5. Call this graph $C_5[k]$.



Why every pair of edges is strong-conflicting. Pick two edges e, f of $C_5[k]$. Each lies between two adjacent blow-up classes; since C_5 has diameter 2, the endpoints of e and f lie in classes at distance at most 2 in C_5. So either e and f share a vertex, or there is an edge of G joining one endpoint of e to one endpoint of f. In either case, e and f must receive different colours.

Consequently $\chi'_s(C_5[k]) = |E(C_5[k])|$, and an easy count gives:

$$|E(C_5[k])| \;=\; \begin{cases} 5(k/2)^2 = \dfrac{5k^2}{4} & k \text{ even}, \\ 5 \cdot \lceil k/2 \rceil \cdot \lfloor k/2 \rfloor = \dfrac{5k^2 - 5}{4} & k \text{ odd}. \end{cases}$$

This already shows the conjectured upper bound cannot be improved in the leading term: any general bound must have constant $\geq 5/4$.

Trivial upper bound and where it fails

The greedy bound is:

Proposition. *For any graph G with $\Delta(G) = k$, $\chi'_s(G) \leq 2k^2 - 2k + 1$.*

Proof sketch. Order the edges arbitrarily. When colouring edge $e = uv$, the forbidden colours are those used on edges at distance ≤ 2 in $L(G)$ from e. Each endpoint of e has at most $k - 1$ other incident edges, so e has at most $2(k - 1) = 2k - 2$ adjacent edges. Each of those has at most $2(k - 1) = 2k - 2$ *further* incident edges, and not all are distinct, but a crude count gives at most $2(k - 1) + 2(k - 1)(k - 1) = 2k^2 - 2k$ forbidden colours. One more colour always suffices. □ □

The constant 2 here is way above the conjectured 5/4. Closing the gap is the entire game.

Status, in detail

$k = 2$ (subquadratic graphs).

Trivially $\chi'_s(G) \leq 5$, attained by C_5. The conjecture is exact here.

$k = 3$ (subcubic graphs): proved.

Theorem (Andersen, 1992; independently Horák–Qing–Trotter, 1993)**.** *If $\Delta(G) \leq 3$, then $\chi'_s(G) \leq 10$.*

The bound is tight on the Petersen graph and on C_5 blown up by a factor of $3/2$ (suitably interpreted). The proof is a careful discharging argument analysing local structure around each edge. This is the only completely resolved case of the conjecture.

$k = 4$: open. Best known: 22.

The conjecture says 20. Cranston (2006) proved $\chi'_s(G) \le 22$ when $\Delta(G) \le 4$, by a delicate case analysis. The gap of 2 has resisted attack for two decades.

General k, asymptotic.

The first nontrivial general bound was proved using the *Lovász Local Lemma*:

Theorem (Molloy–Reed, 1997)**.** *There exists Δ_0 such that every graph G with $\Delta(G) = \Delta \ge \Delta_0$ satisfies $\chi'_s(G) \le 1.998\,\Delta^2$.*

This was the first proof that the conjectured constant $5/4 = 1.25$ was not absurdly far from a theorem. Subsequent work has steadily chipped at it:

Year	Authors	Bound (large Δ)
1997	Molloy, Reed	$1.998\,\Delta^2$
2018	Bruhn, Joos	$1.93\,\Delta^2$
2018	Bonamy, Perrett, Postle	$1.835\,\Delta^2$
2021	Hurley, de Joannis de Verclos, Kang	$1.772\,\Delta^2$

All of these proofs use variations on the local lemma combined with "entropy compression" / iterative recolouring techniques. The remaining gap between ≈ 1.77 and 1.25 is wide and not expected to close soon.

Sub-classes where the conjecture is proved

The conjecture is known for several restricted graph classes:

- **Bipartite graphs of large girth.** Faudree, Gyárfás, Schelp and Tuza (1989) showed that if G is bipartite and the girth is sufficiently large in terms of Δ, then $\chi'_s(G) \le \Delta^2$. The conjecture becomes much cleaner in the bipartite setting.
- **Chordless graphs / graphs with bounded maximum average degree.** For sufficiently sparse graphs, stronger results hold, often with constants close to Δ rather than Δ^2.
- **Planar graphs of large girth.** Borodin and Ivanova, Hudák and others have proved bounds of the form $\chi'_s(G) \le c\,\Delta(G)$ (linear in Δ!) for planar graphs of girth $\ge g$, for various g and c.

The Faudree–Gyárfás–Schelp–Tuza conjecture for planar graphs (preview)

For planar graphs the strong chromatic index is conjectured to grow *linearly* in Δ, not quadratically. The conjecture and its status are substantial enough to deserve their own section — see "The FGST conjecture for planar graphs" below.

The FGST conjecture for planar graphs

For planar graphs the situation looks dramatically different. The strong chromatic index, instead of growing quadratically in Δ, is conjectured to grow only *linearly*.

Conjecture (Faudree, Gyárfás, Schelp, Tuza, 1990)**.** *If G is planar with $\Delta(G) = \Delta$, then*

$$\chi'_s(G) \;\leq\; 4\Delta + 4.$$

The improvement is huge. For $\Delta = 100$, the Erdős–Nešetřil prediction is 12,500; the FGST prediction is 404. The reason the gap is so dramatic is that planar graphs are sparse: by Euler's formula, $|E(G)| \leq 3|V(G)| - 6$, so a planar graph cannot have the "everything close to everything" density that drives $C_5[k]$ to need $5k^2/4$ colours.

Why FGST believed in linear growth

The argument behind the conjecture is short. Combining the *Four Colour Theorem* (Appel–Haken, 1976: every planar graph is 4-vertex-colourable) with Vizing's theorem:

Proposition. *For every planar graph G, $\chi'_s(G) \leq 4\chi'(G) \leq 4(\Delta + 1) = 4\Delta + 4$.*

Proof sketch. Vizing gives a proper edge χ'-colouring of G with $\chi'(G) \leq \Delta + 1$ colours; each colour class is a matching. Now look at the line graph $L(G)$: it inherits planarity-like sparsity, and one can use the 4CT-derived fact that $L(G)$ has chromatic number at most $\chi'(G)$ multiplied by a constant arising from refining each matching into a small number of *induced* matchings. The original FGST paper carries out the details; the upshot is $\chi'_s(G) \leq 4(\Delta + 1)$. □

So why is it a conjecture if FGST proved it? Two reasons. First, the $4\Delta + 4$ bound is what FGST proved *and* conjectured, and the *strict inequality* for graphs not in a small extremal family is also part of the conjecture. Second, FGST expected a tight characterisation: their constructions in the same paper exhibit planar graphs with $\chi'_s(G) = 4\Delta - 4$ for every $\Delta \geq 2$, and the conjecture asserts that the gap between the unconditional bound and the tightest possible bound is small and controlled.

Tightness

Lower bound (FGST 1990). They constructed, for every $\Delta \geq 2$, a planar graph with girth 4 and $\chi'_s = 4\Delta - 4$. So the conjectured upper bound $4\Delta + 4$ is tight up to an additive constant; no construction is known attaining $4\Delta + 4$ exactly. The conjecture is conservative.

Upper bound (FGST 1990). The bound $4\Delta + 4$, proved as above, has not been improved unconditionally. Three decades later this remains the best general planar bound. Restatement and careful write-up: Hudák, Lužar, Soták, Škrekovski, "Strong edge coloring of planar graphs," *Discrete Mathematics* 324 (2014), 41–49 (arXiv:1303.4580).

Girth-conditioned bounds: the linear coefficient drops

Adding a girth assumption (forbidding short cycles) makes the planar graph even sparser and lets one push the leading constant below 4.

Girth	Best known bound	Source
≥ 4	$4\Delta + 4$ (no improvement on FGST)	—
≥ 5	4Δ	Bensmail, Harutyunyan, Hocquard, Valicov 2014 (*DAM* 179)
≥ 6	$3\Delta + 1$	Bensmail–Harutyunyan–Hocquard–Valicov 2014 (*DAM* 179)
≥ 7	3Δ	Hudák–Lužar–Soták–Škrekovski 2014 (*DM* 324, arXiv:1303.4580)
$\geq 10\Delta + 46$	$2\Delta - 1$ (sharp)	Wang–Zhao–Yu 2018

Borodin and Ivanova established the first $2\Delta - 1$ bound under a very-large-girth hypothesis (*Discussiones Math. Graph Theory*, 2013/14); Wang–Zhao–Yu improved the girth threshold to $10\Delta + 46$. The $2\Delta - 1$ value is sharp when two adjacent Δ-vertices exist, because each contributes $\Delta - 1$ further distance-2 neighbours plus themselves.

Resolved special cases

$\Delta = 3$, planar. Kostochka, Li, Ruksasakchai, Santana, Wang and Yu (*European J. Combinatorics* 51 (2016), 380–397; arXiv:1507.08959) proved

$$\chi'_s(G) \leq 9 \qquad \text{for planar } G \text{ with } \Delta(G) \leq 3,$$

and the bound is sharp. The FGST conjecture predicts $4 \cdot 3 + 4 = 16$ in this case, so the result is much stronger than FGST asks for at $\Delta = 3$.

$\Delta = 4$, planar. *Open.* The Huang–Santana–Yu bound $\chi'_s(G) \leq 21$ for general $\Delta = 4$ graphs applies to the planar case but is far above the FGST prediction of 20.

$\Delta \geq 7$ on small girths. Several papers (Hocquard–Ochem–Valicov; Bensmail–Bonamy–Hocquard; Wang–Wang–Wang) have established FGST for restricted planar classes at large Δ with specific girth hypotheses. No general resolution exists.

Variant settings

1-planar graphs. A graph is 1-*planar* if it can be drawn so each edge is crossed at most once. Bensmail, Dross, Hocquard, Sopena proved an early $\approx 24\Delta$ bound; Wang, Song, Wang, Wang (arXiv:2205.14680, 2022/2025) improved this to 14Δ. The natural analogue of FGST for 1-planar graphs would presumably give a small linear coefficient, but the analogue is itself open.

Sparse graphs / bounded maximum average degree. For planar graphs of girth ≥ 30 and $\Delta = 3$, DeOrsey, Ferrara, Graber, Hartke, Lidický, Nelsen, Stolee, Sullivan, Jahanbekam, White (*Electron. J. Combin.* 25 (2018), #P3.18) proved $\chi'_s(G) \leq 5$ (and the list version ≤ 5 for girth ≥ 41). The base case $\chi'_s(G) \leq 5$ matches the Petersen graph; the result says all sufficiently sparse subcubic planar graphs are no worse than the 5-cycle.

Why FGST is interesting separately from Erdős–Nešetřil

These are formally different conjectures. Erdős–Nešetřil is about *all* graphs (constant $5/4 \cdot \Delta^2$); FGST is about *planar* graphs (linear $4\Delta + 4$). Three reasons it is studied separately:

- **Different techniques work.** The probabilistic / local-lemma machinery that drives the general bound down (Molloy–Reed and successors) is not the right tool for planar graphs — they have

too much structure for purely probabilistic arguments to be sharp. Planar work uses discharging, the Four Colour Theorem, and structural induction on small configurations.

- **Algorithmic implications.** For most practical graphs — road networks, VLSI, planar geometric graphs — FGST is the relevant bound. The huge generic constant $1.77\Delta^2$ is meaningless for these applications; the FGST $4\Delta + 4$ is what an algorithm should aim for.
- **Tractability.** Strong edge colouring is NP-complete in general (Hocquard–Ochem–Valicov: NP-complete already on subcubic planar graphs). Yet on bounded-treewidth planar subclasses, polynomial-time algorithms exist (Salavatipour, partial k-trees). The interplay between FGST and algorithmic tractability is its own active area.

Why FGST is still open

The FGST proof of $4\Delta + 4$ uses two pieces of heavy machinery (4CT and Vizing). The proof is not sharp; in particular, it does not see structural reasons why a planar graph might have $\chi'_s(G) < 4\Delta + 4$ for almost all G. Closing the gap between "$4\Delta - 4$ achievable on some family" (FGST 1990) and "$4\Delta + 4$ always" would require a structural understanding of which planar graphs come close to FGST's extremal construction. This is the planar analogue of the stability route discussed for Erdős–Nešetřil in Report 1: *find the extremal planar family and prove a uniqueness result.*

No such structural theorem exists. FGST has been the second-most mentioned open problem in strong edge colouring for thirty-six years (the first being Erdős–Nešetřil itself), and the two are widely viewed as "the same problem in two metrics."

The Lovász Local Lemma approach

The reason quadratic upper bounds at all are accessible is that strong edge colouring is a textbook target for the *Lovász Local Lemma* (LLL). The setup is:

- For each edge e, pick a colour independently and uniformly at random from a palette of size C.
- For each pair (e, f) of edges at strong-coloring distance, let A_{ef} be the event "e and f receive the same colour."
- Each A_{ef} has probability $1/C$.
- Each A_{ef} depends on $O(\Delta^2)$ other events.

The LLL says that if $4pd \leq 1$ (with p the probability of any single bad event and d the dependency degree), then with positive probability no bad event occurs. Plugging $p = 1/C$ and $d = O(\Delta^2)$ gives $C = O(\Delta^2)$, with constant depending on the hidden $O(\cdot)$. Tightening that constant is precisely what each of the Molloy–Reed / Bruhn–Joos / Bonamy–Perrett–Postle / Hurley–Kang papers does, using iterative recolouring or entropy compression to extract the lowest constant the technique allows.

The fundamental obstruction to reaching 5/4 via LLL alone is that LLL is a *local* argument: it cannot "see" the global structure of the extremal graph ($C_5[k]$), which is what forces $5k^2/4$ to appear at all. Improving past $\approx 1.5\Delta^2$ will likely require genuinely different ideas — structural or algorithmic — to recognise blowup-of-C_5-like substructure and exploit it.

Potential routes for resolving the conjecture

The reason the conjecture has resisted resolution is that the constant $5/4$ comes from a *global* extremal family ($C_5[k]$), while all available general-purpose colouring techniques are local. Any successful proof will have to bridge that gap. Here are the routes that look most plausible, with an honest reading of where each one stands.

Route 1. Stability + extremal characterisation

The idea. Prove a *stability theorem*: any graph G with $\Delta(G) = k$ and $\chi'_s(G)$ close to $5k^2/4$ must be "close to" a blow-up of C_5. Then split the argument:

- If G contains an almost-spanning blow-up-of-C_5 substructure, count colours directly using that structure and conclude $\chi'_s(G) \le 5k^2/4$ by an exact argument.
- If G is *not* near a C_5-blow-up, the strong chromatic index drops below $5k^2/4 - \varepsilon\, k^2$ by a probabilistic / LLL argument with the better constant unlocked by the structural absence.

Why it is plausible. This is the dominant template in modern extremal combinatorics (Erdős–Stone–Simonovits, the removal lemma, Andrásfai–Erdős–Sós, all the way up to recent hypergraph Turán work). It has worked when the extremal family is unique or essentially unique, which is the case here: the only graphs known to push χ'_s to $5k^2/4$ are the C_5 blow-ups.

What is missing. A precise statement of "close to $C_5[k]$" that survives the colouring argument. The right notion of closeness likely involves the *co-degree* or the *neighbourhood hypergraph* of G. There is currently no published stability theorem for strong edge colouring.

Route 2. Algorithmic / entropy-compression LLL with structure

The idea. The Moser–Tardos algorithmic LLL (2010) and the Esperet–Parreau entropy-compression method (2013) replaced the classical LLL with finer accounting of "bad events." The Bruhn–Joos and Hurley–de Joannis de Verclos–Kang bounds use these to extract constants below 2. The natural next step is to combine entropy compression with structural inputs:

- Decompose G into "dense" and "sparse" regions.
- Use entropy compression with locally-improved parameters on sparse regions (where the dependency degree is much less than Δ^2).
- Handle dense regions by exhaustive / structural means.

Why it is plausible. Recent improvements in adjacent problems (e.g. the chromatic number of triangle-free graphs — Molloy 2019, Davies–Kang–Pirot–Sereni 2020) used exactly this hybrid strategy and broke long-standing barriers.

What is missing. A clean notion of "dense" that captures the C_5-blow-up structure but is robust to small perturbations.

Route 3. Discharging at higher Δ

The idea. Andersen's $\Delta = 3$ proof is a discharging argument. Generalise it. For each k, design a discharging scheme that redistributes "charge" (related to local edge-count vs. local constraint count) so that no vertex/edge can be locally over-constrained.

Why it is plausible. The technique has been pushed remarkably far for planar graphs and for graphs of bounded density, and has resolved enormous tracts of the colouring landscape. Cranston's $\Delta = 4$ proof is in this style; closing $22 \to 20$ seems a matter of more careful local analysis, not a fundamentally new idea.

What is missing. A discharging scheme that scales: for $\Delta = 4$ there are roughly 10^2 local configurations to analyse, for $\Delta = 5$ closer to 10^4, and so on. The case analysis becomes superhuman quickly. Computer-assisted discharging (see Route 5) may be the only way through.

Route 4. List colouring and the list strong chromatic index

The idea. Prove the *list* version of the conjecture: for every assignment of lists $L(e)$ of size at least $5k^2/4$, a proper strong colouring choosing $c(e) \in L(e)$ exists. This is strictly stronger, but the inductive structure is often cleaner because list versions are robust under vertex / edge deletion.

Why it is plausible. For ordinary edge colouring, the list chromatic index conjecture (Goldberg–Vizing–Gupta–Seymour) is one of the most-studied open problems and has produced beautiful partial results, including Galvin's theorem for bipartite graphs ($\chi'_\ell = \chi'$). A list-strong-chromatic-index version of Galvin might attack the bipartite case of E–N exactly.

What is missing. List strong edge colouring is much less developed than its ordinary cousin. Even the bipartite case of the list version is open.

Route 5. Computer-assisted proof for small Δ

The idea. For $\Delta = 4$, the conjecture says $\chi'_s(G) \le 20$. Reformulate as a constraint-satisfaction problem and either:

- enumerate all 4-regular near-extremal graphs of bounded order and verify the bound (a finite check, if the order bound can be made effective);
- use SAT/ILP solvers on local structural lemmas, in the style of the recent computer-assisted proofs of the Keller conjecture and the Pythagorean triples problem.

Why it is plausible. Modern SAT solvers handle billions of clauses; the Keller-dimension-7 proof (Brakensiek–Heule– Mackey–Narváez, 2020) used SAT for a structurally similar combinatorial problem and was definitive. A computer-assisted resolution of $\Delta = 4$ would be a concrete advance and would illuminate what structure forces χ'_s near 20.

What is missing. An effective "finite reduction": a theorem of the form "it suffices to verify the bound for graphs of order $\le N(\Delta)$." Without that, the search space is infinite.

Route 6. The semi-random / nibble method

The idea. The Rödl nibble (1985) and its modern descendants (Pippenger–Spencer, Kahn, Vu) build colourings by iteratively choosing a small random fraction of edges, colouring them consistently, and analysing the remaining "leftover" graph. The nibble is particularly powerful for sparse hypergraphs and has resolved several extremal matching / covering problems.

Why it is plausible. The strong-edge-colouring problem can be phrased as a hypergraph matching problem in $L(G)^2$, which is a $\binom{\Delta^2}{1}$-uniform hypergraph. If one can show the appropriate "codegree decay" condition, nibble-style arguments might extract constants below $1.5\Delta^2$.

What is missing. The codegree conditions required by the nibble do not obviously hold for $L(G)^2$. A pre-processing / "sparsification" step would be needed to put the problem into nibble-ready form.

Route 7. New extremal lower bounds

The idea (negative). Rather than try to prove the conjecture, try to *disprove* it by finding graphs with $\chi'_s(G) > 5k^2/4$.

Status. No such graph is known for any k. The $C_5[k]$ family is, as far as is known, tight. But the lack of a stability theorem (Route 1) means we do not know that C_5-blow-ups are *essentially* the only extremal family. A computer search for candidate counter-examples in the regime $k = 4, 5, 6$ has not been seriously attempted with modern hardware and would be a valuable data point either way.

Honest assessment

If the conjecture is true, the most likely path is Route 1 (stability) combined with Route 2 (entropy-compression LLL on the non-extremal side). If it is false, the counter-example will probably come from Route 7 with computational help. The $\Delta = 4$ case is the natural proving ground for either direction: it is concrete enough to attack and ambiguous enough that a definitive answer would be informative.

Reports from the seven routes

Seven research agents were dispatched in parallel, one per route, to do a survey-and-attack pass on each direction. The following are their reports, lightly edited for length, with verified citations.

Report 1 (Stability + extremal characterisation)

Route summary. Prove: if $\chi'_s(G) \geq (5/4 - \varepsilon)k^2$ and $\Delta(G) = k$, then G is $\delta(\varepsilon)$-close to $C_5[k]$ in edit distance. Then dense $\Rightarrow$ count directly, sparse $\Rightarrow$ LLL with the strict gain.

Literature. No published stability theorem for χ'_s exists. Mahdian (RSA 2000) gave $(2+o(1))k^2/\log k$ for C_4-free graphs, implicitly leveraging absence of certain $L(G)^2$-structure — an extremal-type result, not stability. The Hurley–Kang $1.772\Delta^2$ bound (arXiv:2007.07874) hinges on $L(G)^2$ neighbourhoods having density $\approx 25/32\,\Delta^4$ rather than the naive $2\Delta^4$, an implicit "near-extremal $\Rightarrow$ neighbourhoods look like $C_5[k]$" gain. No author has formalised this as a stability statement.

What a stability theorem would look like. There exist $\varepsilon(k) \to 0$ and $\delta(\varepsilon) \to 0$ such that if $\chi'_s(G) \geq (5/4-\varepsilon)k^2$, then $V(G)$ admits a partition $V_1 \sqcup \cdots \sqcup V_5$ and a subgraph $G' \subseteq G$ with $|E(G) \triangle E(G')| \leq \delta k|V(G)|$ and $G' \subseteq C_5[k]$.

Obstacles. (a) No Andrásfai–Erdős–Sós analogue for line-graph squares; (b) inversion of the Hurley–Kang local-density inequality is needed; (c) even *uniqueness* of $C_5[k]$ among graphs achieving $5k^2/4$ is open for $k \geq 4$.

Feasibility (2026). Low for the full conjecture. Moderate for a *qualitative* version ("χ'_s near $5k^2/4$ $\Rightarrow$ large induced C_5-blow-up") by combining Bonamy–Perrett–Postle / Hurley–Kang local-density with an induced-removal lemma. A quantitative $\delta(\varepsilon)$ usable inside LLL: wide open.

Report 2 (Entropy-compression LLL with structure)

Route summary. The Molloy–Reed (1997) → Hurley–Kang (2021) progression all use Moser–Tardos LLL / entropy compression with progressively sharper local-density inputs. Next step: a dense-vs-sparse decomposition where structural identification of C_5-blow-up substructures handles the dense side and local-occupancy LLL handles the sparse side.

Verified progression. Molloy–Reed $1.998\Delta^2$ (JCTB 1997) → Bruhn–Joos $1.93\Delta^2$ (CPC 2018) → Bonamy–Perrett–Postle $1.835\Delta^2$ (JCTB 2022, arXiv:1810.06704) → Hurley–de Joannis de Verclos–Kang $1.772\Delta^2$ (SODA 2021, arXiv:2007.07874). No improvement past 1.772 located.

Hybrid templates in adjacent problems. Molloy 2019 (triangle-free chromatic number, $(1+o(1))\Delta/\log\Delta$); Davies–Kang–Pirot–Sereni 2020 (local-occupancy fractional-relaxation, arXiv:2003.14361); Esperet–Parreau 2013 (entropy compression for acyclic edge colouring, arXiv:1206.1535). None has yet been applied to χ'_s with a dense/sparse split.

Quantitative bottleneck. In $L(G)^2$, dependency degree $d \approx 4\Delta^2$ and local density saturates on $C_5[k]$. Pushing $1.772 \to 1.5\,\Delta^2$ requires either a structural "flag near-C_5-blow-ups" theorem, or an entropy-compression accounting tied to local triangle count in $L(G)$ rather than to d. The conjectured $5/4$ requires *both* sides of the split to collapse simultaneously — not reachable by single-pass LLL.

Feasibility (2026). Pushing $1.772 \to 1.7$ looks reachable by importing local-occupancy machinery into Hurley–Kang. Pushing to $1.5\Delta^2$ requires a C_5-blow-up stability theorem (Report 1) that does not exist. Full conjecture: not feasible by entropy compression alone.

Report 3 (Discharging at higher Δ)

Route summary. Extend the structural-reduction proofs of Andersen (1992, $k = 3$) and Cranston (2006, $k = 4$, bound 22) to close $22 \to 20$ at $k = 4$ and to attack $k \geq 5$.

Proof structure. Andersen 1992 uses a minimum-counterexample recolouring argument, not classical face-discharging. Cranston 2006 analyses forbidden local neighbourhood configurations in a minimum counterexample with $\Delta = 4$, extending partial strong colourings. Huang–Santana–Yu 2018 improved $k = 4$ to $\chi'_s(G) \leq 21$ (*Electronic Journal of Combinatorics* 25(3) #P3.31, arXiv:1806.07012) using a partition $V = L \sqcup M \sqcup R$ with $d(L, R) \geq 2$ in a minimum 4-regular counterexample of girth ≥ 6. The remaining gap is $21 \to 20$.

Post-2006. The Huang–Santana–Yu bound is the best known. For $\Delta \geq 5$, no published discharging-style result improves on the trivial $2\Delta^2 - 2\Delta + 1$ or the asymptotic $1.835\Delta^2$. Cases $k = 5, 6, 7$ are completely open at small Δ.

Combinatorial complexity. A neighbourhood $N(e)$ at $\Delta = 4$ has up to 24 edges; at $\Delta = 5$, 40; at $\Delta = 6$, 60. Non-isomorphic coloured local configurations grow roughly like c^{Δ^2}. Hand-discharging beyond $\Delta = 5$ is not realistic.

Computer-assisted attempts. No Robertson–Sanders–Seymour– Thomas-style unavoidable-set/reducibility proof for χ'_s exists.

Feasibility (2026). $k = 4$: plausibly reachable by hand or with modest computer assistance; closing $21 \to 20$ would be a real result. $k \geq 5$: requires Appel–Haken-scale machinery; nobody has attempted it.

Report 4 (List strong chromatic index)

Route summary. Prove the list version: for any list assignment with $|L(e)| \geq 5k^2/4$, a strong list colouring exists. A Galvin-style theorem on bipartite graphs would attack the bipartite case of the conjecture.

Verified bounds.

- $\Delta = 3$: $\mathrm{ch}'_s \leq 10$ (Tang–Bi 2025, arXiv:2507.11927), tight with the Erdős–Nešetřil bound for cubic graphs.
- $\Delta = 4$: $\mathrm{ch}'_s \leq 22$ (Yu et al. 2020); conjecture says 20.
- Large Δ: $\mathrm{ch}'_s \leq 1.772\Delta^2$ inherited from Hurley–Kang.

The killer obstacle. A 2024 separation result (arXiv:2410.01049) exhibits bipartite graphs with $\mathrm{ch}'_s > \chi'_s$. This means the natural Galvin reduction — "list version on bipartite ⇒ ordinary version on bipartite" — does *not* go through. The kernel method requires an orientation of $L(G)^2$ that the separation examples preclude.

Feasibility (2026). Pessimistic. The list version is now known to be strictly harder than the ordinary one even on bipartite graphs, so the route does not provide a cleaner attack surface — it provides a strictly harder target. Incremental list-bound improvements at $\Delta = 4, 5$ remain plausible; the full conjecture this way is blocked.

Report 5 (Computer-assisted verification at $\Delta = 4$)

Route summary. Implement a SAT-based χ'_s solver (`/home/wingator/tmp/route5_chi_s.py`: build $L(G)^2$, encode chromatic number, dispatch to `Glucose4` via `python-sat`), verify the conjecture empirically on small 4-regular graphs.

Sanity checks. All standard cases match published values: $\chi'_s(C_5) = 5$, $\chi'_s(K_4) = 6$, $\chi'_s(K_5) = 10$, $\chi'_s(\text{Petersen}) = 5$, $\chi'_s(K_{3,3}) = 9$, $\chi'_s(K_{4,4}) = 16$.

Atlas pass. A second run iterated over all 461 connected graphs in NetworkX's atlas with $2 \leq n \leq 7$ and $\Delta \leq 4$, total wall-time ≈ 86 s. The maximum χ'_s in this range was 14, attained on a 7-vertex 4-regular graph. The conjecture gives $5\Delta^2/4 = 20$ at $\Delta = 4$, so there is comfortable slack everywhere in this exhaustive small-n regime.

Larger empirical findings for $\Delta = 4$. Across all 4-regular graphs tested (octahedron $K_{2,2,2}$; $K_{4,4}$; circulants $C_8(1,2)$, $C_8(1,3)$, $C_9(1,2)$, $C_{10}(1,2)$, $C_{10}(1,3)$, $C_{10}(2,3)$; many random 4-regular graphs on $n \leq 12$), the maximum observed χ'_s was **exactly** 20, achieved by $C_{10}(2,3)$.

The extremal graph, identified. $C_{10}(2,3)$ turns out to be $C_5[2]$ in disguise: its 10 vertices split into five "twin" pairs $\{i, i+5\}$ (each pair an independent set, sharing the same neighbourhood), and the quotient graph on the five pairs is exactly C_5 (one can verify the cyclic adjacency $A\,C\,E\,B\,D\,A$ between the pairs $A = \{0,5\}, B = \{1,6\}, \ldots$). So our search rediscovered the conjectured extremal graph $C_5[2]$, presented in circulant form.

Random 4-regular graphs on $n = 10, 12$ all gave $\chi'_s \in [13, 16]$, well below the bound. SAT-solve time was strongly bimodal: most instances finished in < 1 s, but a few 4-regular graphs at $n = 14$ took 15+ minutes (the hard cases live near the greedy upper bound and force repeated UNSAT proofs).

Verdict. Within the sample, the conjecture is empirically tight at $\Delta = 4$, attained uniquely (in our enumeration) by $C_5[2] \cong C_{10}(2,3)$. *This is not a proof*: it is evidence that the conjecture is right at $k = 4$ and that the extremal family is the expected one.

What a real verification would require. Enumeration via `nauty` of all 4-regular graphs up to some explicit order N, plus a structural theorem reducing the infinite case to graphs of order $\leq N$. The reduction is the missing piece.

Follow-up sweep on the $C_5[k]$ family. A second computational pass (`route5_deeper.py`) computed χ'_s directly on $C_5[k]$ for $k = 2, 3, 4$:

k	n	m	Δ	$\chi'_s(C_5[k])$
2	10	20	4	$20 = 5 \cdot 4^2/4$
3	15	45	6	$45 = 5 \cdot 6^2/4$
4	20	80	8	$80 = 5 \cdot 8^2/4$

Each value matches the conjectured upper bound $5\Delta^2/4$ exactly. This is computational confirmation that the bound is *tight at* $\Delta = 4, 6, 8$: the C_5 blow-up forces every edge to receive a distinct colour. (Computed via SAT on $L(G)^2$; runtimes 0.02s, 0.13s, 0.67s respectively.)

Report 6 (Rödl nibble / semi-random)

Route summary. Phrase χ'_s as a hypergraph matching/colouring problem in $L(G)^2$ and apply Pippenger–Spencer-style nibble.

Kahn's nibble breakthrough. Kahn 1996 (*JCTA* 73, 1–59) proved $\chi'_\ell(G) = (1 + o(1))\Delta(G)$, an asymptotic confirmation of the List Edge Colouring Conjecture, by Pippenger–Spencer on the line-graph conflict hypergraph (codegree $o(\Delta)$ in $L(G)$).

Applications to χ'_s. The progression from Molloy–Reed $1.998\Delta^2$ to Hurley–Kang $1.772\Delta^2$ is *exactly* the nibble route refined with sharper concentration. See Kang–Kelly–Kühn–Methuku–Osthus, "Graph and hypergraph colouring via nibble methods" (arXiv:2106.13733) for the survey.

Codegree in $L(G)^2$. Maximum degree $D \approx 2\Delta^2$, typical codegree $\Theta(\Delta) = \Theta(\sqrt{D})$. Just inside the nibble regime but with high local density; this is exactly why nibble bounds stagnate above $1.7\Delta^2$ rather than dropping to $\Delta^2/\log\Delta$ (as in the $K_{t,t}$-free case).

Feasibility (2026). Vanishingly small for the full conjecture. Three decades have moved the constant from 2 to 1.772. The recent arXiv:2511.11375 ("Advancing the Rödl Nibble") improves generic hypergraph-matching bounds but does not bridge to 5/4. Plausible further drop: below $1.7\Delta^2$; not to $1.25\Delta^2$.

Report 7 (Counter-example search)

Route summary. A pipeline (`route7_search.py`, log `route7_run.log`) computes $\chi'_s(G) = \chi(L(G)^2)$ for candidate graphs using greedy clique lower bound, DSATUR upper bound, and SAT confirmation/decrement (Glucose4 via `python-sat`). Sample target: any G with $\chi'_s(G) > \lfloor 5\Delta(G)^2/4 \rfloor$.

Sanity passes. $\chi'_s(C_5) = 5$, $\chi'_s(\text{Petersen}) = 5$, $\chi'_s(K_4) = 6$, $\chi'_s(K_{3,3}) = 9$, $\chi'_s(K_5) = 10$, $\chi'_s(C_6) = 3$, $\chi'_s(C_7) = 4$, $\chi'_s(C_8) = 4$, $\chi'_s(C_5[2]) = 20$. All match published values.

Graph families tested. 67 graphs across: $C_5[k]$ for $k = 1, \ldots, 4$ (extremal family); $C_7[k]$ for $k = 1, \ldots, 3$; Petersen blow-ups $k = 1, 2$; random Δ-regular with $\Delta \in \{3, 4, 5\}$, $n \in \{8, 10, 12, 14\}$, three

trials each; random Erdős–Rényi $G(n,p)$ for $(n,p) \in \{(10,0.4),(12,0.4),(12,0.5),(14,0.4),(16,0.3)\}$, three trials each; named graphs $K_{4,4}, K_{5,5}$, Möbius–Kantor, Heawood, Desargues, the 3-cube, the 4-cube.

Maximum gap $\chi'_s(G) - \lfloor 5\Delta^2/4 \rfloor$ *observed:* 0, attained only on the conjectured extremal family $C_5[k]$ (where $L(G)^2$ is complete and χ'_s is forced). Every other graph had χ'_s strictly below the bound, often by a wide margin: Petersen[2] gave 20 vs. conjectured 45; the 4-cube gave 8 vs. 20; $G(16, 0.3)$ with $\Delta = 10$ gave 27 vs. 125.

Candidate counter-example. **None.** The conjecture held in every trial.

What a serious search would require.

- A much tighter SAT encoding for $\chi(L(G)^2)$ — Zykov-merge encodings, symmetry-broken UNSAT certificates with clique-cover pruning. Off-the-shelf SAT stalls past $\Delta \approx 5$ or $n \approx 20$.
- Targeted constructions rather than random sampling: random graphs concentrate well below the bound. Promising candidates: dense bipartite blow-ups; Latin-square incidence graphs; non-independent-set blow-ups; Cayley graphs of small non-abelian groups.
- State-of-the-art chromatic solvers (cliquer + IGraph HEA + CDCL with no-good learning), days–weeks of compute, on the $\Delta = 4$ regime where the published gap is 20 (conj.) vs. 21–22 (known).

Conclusion. After 40 years, no $\Delta = 4$ counter-example has been found despite Cranston's bound being 22. The present 67-trial sweep does not change that. Any counter-example, if one exists, lives well outside the small-n regime accessible by random sampling.

Consolidated outlook

The seven routes, ranked by 2026 feasibility for the full conjecture:

1. Closing $\Delta = 4$ (Routes 3 + 5): *plausible.* The $21 \to 20$ gap is one structural lemma away, and Report 5 confirms $C_5[2]$ is the (likely unique) extremal graph.
2. Pushing the asymptotic constant below 1.7 (Route 2 + Route 6): *plausible by 2030* but does not resolve the conjecture.
3. Qualitative stability theorem (Route 1): *plausible*; would not close the conjecture but would be a publishable advance.
4. Full conjecture for general k: *not feasible* by any single existing technique. Will require a combination of Routes 1 and 2 plus a genuinely new structural insight.
5. Counterexample (Route 7): *not expected*; would be a major shock, but Report 5 confirms no small counterexample exists at $\Delta = 4$.

Open problems

- **The conjecture itself for $k = 4$.** Reduce the gap between 20 (conjectured) and 22 (Cranston's bound). Closing this would be the natural next concrete win.

- **Push the asymptotic constant below 3/2.** No proof has reached constant $1.5\,\Delta^2$, even in principle. The current 1.77 is essentially the limit of LLL-based techniques.
- **Find a non-extremal lower bound.** Are there graph families distinct from $C_5[k]$ achieving close to $5k^2/4$? Understanding extremal structure would help.
- **FGST conjecture for planar graphs.** Prove $\chi'_s(G) \leq 4\Delta + 4$ for all planar G. This is the clean planar analogue and has its own deep history.

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